

# Enterprise Vault Deployment on AWS – Architecture Design Document

---

## 1. Executive Summary

The objective of this project is to deploy Veritas Enterprise Vault on AWS to provide a highly available, secure, and scalable archiving platform for email, file systems, and enterprise content.

The AWS solution will leverage managed services wherever possible to reduce operational overhead while maintaining Enterprise Vault application requirements.

---

## 2. Business Requirements

### Functional Requirements

- Archive Exchange mailboxes
- Archive File Shares
- Store archived data securely
- Support retention and compliance requirements
- Enable Enterprise Vault Search
- Provide disaster recovery capabilities

### Non-Functional Requirements

- High Availability
  - Scalability
  - Data Durability
  - Security and Compliance
  - Backup and Recovery
  - Monitoring and Alerting
- 

## 3. Target AWS Architecture

### AWS Region

Primary Region:

ap-south-1 (Mumbai)

Disaster Recovery Region:

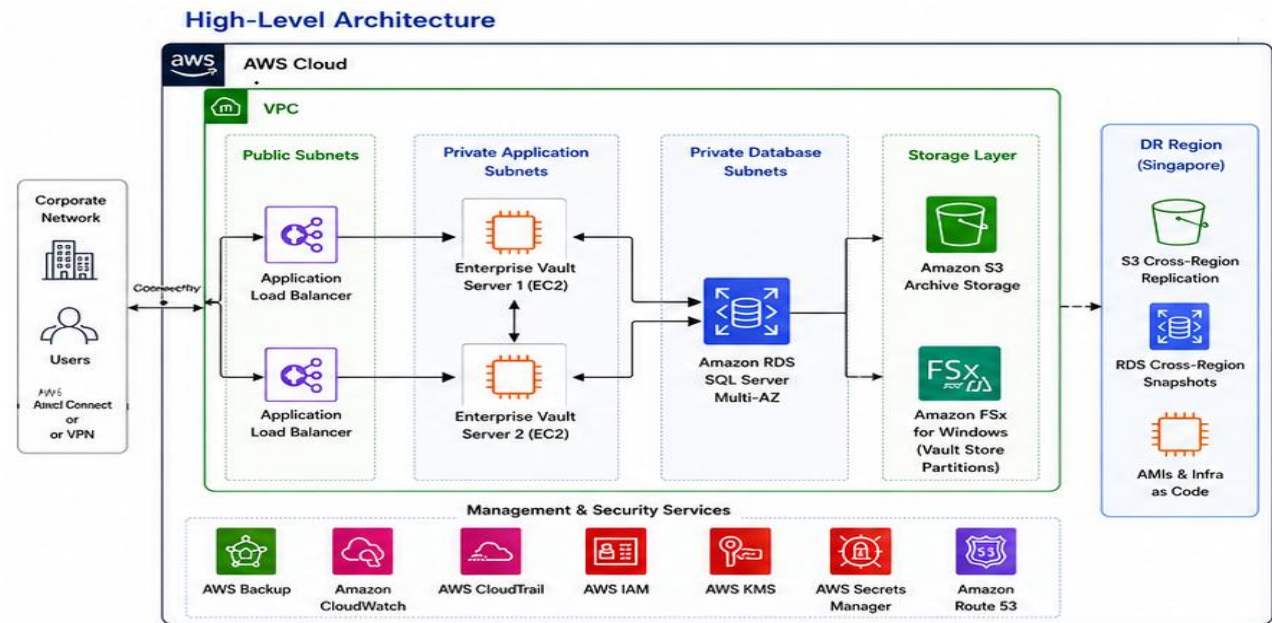
ap-southeast-1 (Singapore)

---

## Core AWS Services

| Component      | AWS Service                 |
|----------------|-----------------------------|
| Networking     | VPC                         |
| Compute        | EC2                         |
| Database       | Amazon RDS SQL Server       |
| Storage        | Amazon S3                   |
| File Storage   | Amazon FSx for Windows      |
| Backup         | AWS Backup                  |
| Monitoring     | CloudWatch                  |
| Logging        | CloudTrail                  |
| Security       | IAM, KMS, Secrets Manager   |
| DR Replication | S3 Cross Region Replication |
| DNS            | Route53                     |
| Connectivity   | Direct Connect / VPN        |

## 4. High-Level Architecture



## 5. Network Architecture

### VPC Design

CIDR: 10.0.0.0/16

#### Subnets

| Subnet           | CIDR         | Purpose    |
|------------------|--------------|------------|
| Public AZ-A      | 10.0.1.0/24  | ALB        |
| Public AZ-B      | 10.0.2.0/24  | ALB        |
| Private App AZ-A | 10.0.11.0/24 | EV Servers |
| Private App AZ-B | 10.0.12.0/24 | EV Servers |
| DB AZ-A          | 10.0.21.0/24 | RDS        |
| DB AZ-B          | 10.0.22.0/24 | RDS        |

## 6. Enterprise Vault Components

### EV Application Servers

#### Configuration

Instance Type: m7i.xlarge

Operating System: Windows Server 2022

Quantity: 2

Functions:

- Storage Service
- Task Controller
- Indexing Service
- Admin service
- Directory Service

Deployment:

- Private Subnets
  - Auto Recovery Enabled
  - Multi-AZ Placement
- 

### Load Balancer

Application Load Balancer Functions:

- Load Distribution
  - SSL Offloading
  - Health Checks
- 

## 7. Automated EC2 Provisioning Using User Data

It is recommended creating a golden AMI with Windows Server, .NET Framework, IIS, MSMQ, and Enterprise Vault prerequisites already installed. Then the User Data only performs:

1. Domain join
2. Configuration retrieval from AWS Secrets Manager
3. Enterprise Vault installation
4. Registration with monitoring tools

#### EC2 Automation Approach:

User Data responsible for:

- Initial bootstrapping
- Domain Join
- SSM Agent validation
- Logging setup

#### AWS Systems Manager State Manager

Responsible for:

- Enterprise Vault installation
- Windows feature installation
- SQL connectivity validation
- Post-install configuration

### Objective

Automate deployment of Enterprise Vault servers during EC2 launch by:

- Joining the server to Active Directory
- Installing required Windows features
- Installing Enterprise Vault prerequisites
- Installing Enterprise Vault binaries
- Configuring services
- Registering with monitoring tools

### Prerequisites

#### AWS Components

- AWS Managed Microsoft AD or On-Prem AD via Direct Connect/VPN
- Secrets Manager storing:
  - Domain Join Account
  - Enterprise Vault Service Account
  - SQL Service Account

### IAM Role

Attach an IAM Role to Enterprise Vault EC2 instances with permissions:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetSecretValue"
      ],
      "Resource": "*"
    }
  ]
}
```

### EC2 User Data Script to join the EC2 instance to domain:

```
# -----
# Variables
# -----
$DomainName = "corp.company.com"
$DomainOU = "OU=EnterpriseVault,OU=Servers,DC=corp,DC=company,DC=com"
# Retrieve credentials from Secrets Manager
```

```

$Secret = (Get-SECSecretValue `
    -SecretId "DomainJoinCredentials").SecretString
$Creds = $Secret | ConvertFrom-Json
$User = $Creds.username
$Pass = $Creds.password
$SecurePass = ConvertTo-SecureString $Pass -AsPlainText -Force
$Credential = New-Object `
    System.Management.Automation.PSCredential `
    ($User,$SecurePass)
# -----
# Rename Computer
# -----
$InstanceID = Invoke-RestMethod `
    http://169.254.169.254/latest/meta-data/instance-id
Rename-Computer -NewName "EV-$(($InstanceID.Substring(0,6)))"
# -----
# Join Domain
# -----
Add-Computer `
    -DomainName $DomainName `
    -Credential $Credential `
    -OUPath $DomainOU `
    -Restart

```

### Post-Domain Join Configuration

After reboot, use AWS Systems Manager Automation or Scheduled Tasks.

```

# Install IIS
Install-WindowsFeature `
    Web-Server `
    -Web-Asp-Net45 `
    -Web-Mgmt-Tools
# MSMQ
Install-WindowsFeature `
    MSMQ-Server
# .NET Framework
Install-WindowsFeature `
    NET-Framework-45-Core
Install-WindowsFeature `
    RSAT-AD-PowerShell

```

### Enterprise Vault Silent Installation

Store installation media on:

```

s3://enterprise-vault-installers/
Enterprise Vault Installation Script

```

```

$EVMedia = "s3://enterprise-vault-installers/"
Start-Process `
    "$EVMedia\setup (x64).exe" `

```

```
-ArgumentList `
"/s /clone_wait /v"LOGFILE="D:\Logs\EVinstall.log" COMPONENTS=VAULT" `
-Wait
```

#### **SQL Connectivity Validation**

```
Test-NetConnection `
-SQLSERVER.company.local `
-Port 1433
```

---

## **8. Database Architecture**

### **Amazon RDS SQL Server**

Edition: SQL Server Enterprise

Deployment: Multi-AZ

Storage: GP3 SSD

Initial Size: 500 GB Backup

Retention: 35 Days Databases:

- EnterpriseVaultDirectory
  - EnterpriseVaultMonitoring
  - Vault Store Databases
  - Fingerprint Database Benefits:
    - Automated Backups
    - Automated Patching
    - Multi-AZ Failover
- 

## **9. Storage Architecture**

### **Option 1 – Recommended**

#### **Amazon S3 (Cheapest)**

Purpose: Archive Storage

Storage Classes:

- S3 Standard

- S3 Intelligent Tiering
  - S3 Glacier Deep Archive Benefits:
  - 11 9's Durability
  - Lifecycle Management
  - Cost Optimization
- 

## Option 2

### Amazon FSx for Windows

Purpose: Vault Store Partitions Features:

- SMB Compatible
  - Active Directory Integration
  - Backup Integration
- 

## 10. Security Architecture

### Identity Management

Microsoft Active Directory

Integrated with:

- AWS Managed Microsoft AD
  - Enterprise Vault Services
- 

### Encryption

#### Data at Rest

- AWS KMS
- RDS Encryption
- EBS Encryption
- S3 Encryption

#### Data in Transit

- TLS 1.2+
  - HTTPS
  - SMB Encryption
-

## Secrets Management

AWS Secrets Manager Stores:

- SQL Credentials
- Service Accounts
- API Keys

---

## 11. High Availability Design

### Application Layer

Enterprise Vault Servers

- Multiple EC2 Instances
- Separate Availability Zones

### Database Layer

RDS SQL Server Multi-AZ

Automatic Failover

### Storage Layer

Amazon S3

99.999999999% Durability

---

## 12. Backup Strategy

### AWS Backup

Protected Resources:

- EC2
- EBS
- FSx

RDS Retention:

Daily: 30 Days

Weekly: 12 Weeks

Monthly: 12 Months

Yearly: 7 Years

## 13. Disaster Recovery

### DR Region

Singapore Region

#### Replicated Components

- S3 Archives
- RDS Snapshots
- AMIs
- Infrastructure as Code Templates

#### Target RTO

4 Hours

#### Target RPO

15 Minutes

---

## 14. Monitoring and Logging

### Amazon CloudWatch

Monitors:

- CPU
- Memory
- Disk
- Network
- Enterprise Vault Services

### AWS CloudTrail

Tracks:

- User Activity
- API Calls
- Security Events

### AWS Config

Tracks:

- Resource Changes
  - Compliance
-

## 15. Cost Optimization

### Recommendations

- Use GP3 volumes
- S3 Intelligent Tiering
- Reserved Instances for EV Servers
- Savings Plans
- Lifecycle Policies to Glacier

Expected Cost Savings: 30–40% compared to traditional datacenter infrastructure.

---

## 16. Deployment Approach

Phase 1: Assessment and Discovery

Phase 2: AWS Landing Zone Setup

Phase 3: Network Connectivity Setup Phase 4:

Database Deployment

Phase 5: Enterprise Vault Installation

Phase 6: Archive Migration

Phase 7: Validation and Testing

Phase 8: Production Cutover

---

## 17. Architecture Benefits

- Highly Available
  - Secure
  - Scalable
  - Cost Optimized
  - Disaster Recovery Ready
  - Compliance Ready
  - Reduced Operational Overhead
-

## 18. Conclusion

The proposed AWS architecture modernizes Enterprise Vault by leveraging AWS managed services, Multi-AZ deployment, automated backups, and scalable archive storage. The solution provides enterprise-grade security, resilience, and operational efficiency while supporting future growth.